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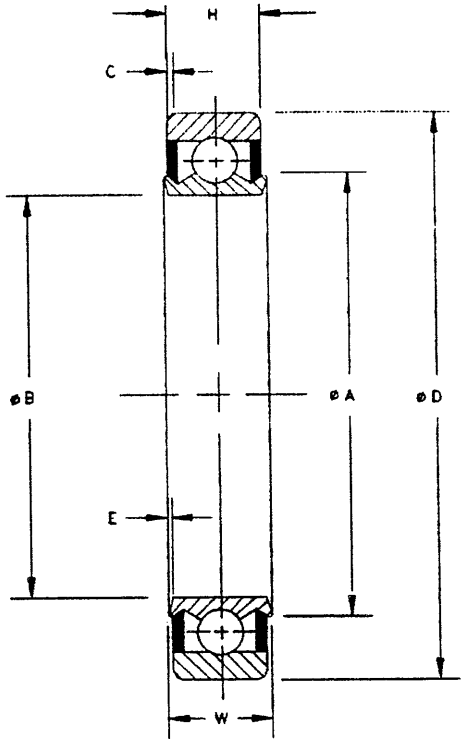


TABLE I. DIMENSIONS AND LOADS

MS DASH NO.	ØB 2/ BORE	BORE TOLERANCE	ØD 1/ 2/ OUTSIDE DIAMETER	OUTSIDE DIAMETER TOLERANCE	W 1/ WIDTH INNER RING +.0000 −.0025	H 1/ WIDTH OUTER RING +.000 −.005	ØA SHOULDER DIAMETER INNER RING APPROX.	E & C 3/ 4/ CORNER CHAMFER 45° +.020 −.000	WEIGHT POUNDS (APPROX)	RADIAL LIMIT LOAD RATING LBS	THRUST LIMIT LOAD RATING LBS	5/ 6/ RADIAL LOAD RATING (LBS.) FOR AVERAGE LIFE OF 10,000 COMPLETE 90° CYCLES CASE I	CASE II	MAXIMUM STARTING TORQUE (INCH-OZ)
-38	.6250		1.0625				.777		.03	3280	1500	1990	1820	2
-39	.7500		1.1875				.895		.04	3750	1700	2050	1900	2
-40	.8750	+.0000	1.3125	+.0000			1.016		.05	4220	1900	2110	1970	3
-41	1.0625	−.0005	1.5000	−.0005	.2810	.250	1.216	.015	.06	5000	2200	2170	2020	4
-42	1.3125		1.7500				1.451		.09	5950	2700	2220	2130	4
-43	1.5625		2.0000				1.702		.10	6880	3200	2260	2180	5
-44	1.8125	+.0000	2.2500	+.0000			1.970		.11	7960	3600	2300	2220	5
-45	2.0625	−.0008	2.6250	−.0007			2.286		.15	9220	4000	2340	2260	6
-46	2.3125		2.8750				2.527		.17	10150	4400	2360	2280	6

1/ DIMENSIONS TO BE MET AFTER PLATING.
2/ OUT-OF-ROUND TOLERANCES:

BORE:
FOR −38 THRU −43: +.0003, −.0008
FOR −44 THRU −46: +.0003, −.0011

OUTER DIAMETER:
FOR −38 THRU −43: +.0005, −.0010
FOR −44 THRU −46: +.0005, −.0012

3/ A RADIUS GIVING APPROXIMATELY THE SAME GRIP FOR STAKING THE BEARING AND MAINTAINING DIMENSION C IS ACCEPTABLE IN LIEU OF THE 45° FLAT SURFACE CHAMFER.
4/ A RADIUS GIVING APPROXIMATELY THE SAME FILLET CLEARANCE AND MAINTAINING DIMENSION E IS ACCEPTABLE IN LIEU OF THE 45° FLAT SURFACE CHAMFER.
5/ CASE I = LOAD IS FIXED WITH RESPECT TO OUTER RING.
CASE II = LOAD IS FIXED WITH RESPECT TO THE INNER RING.
6/ THESE RATINGS ARE FOR OPERATION UP TO 250°F FOR OPERATION UP TO 350°F, THE RATINGS SHALL BE REDUCED BY 20%.

Ⓒ DENOTES CHANGES

INCH-POUND

PREPARING ACTIVITY: NAVY-AS
CUSTODIANS: ARMY- AV NAVY- AS
AIR FORCE- 99 DLA-
REVIEWARMY-AR,AT,MI AIR FORCE-84, DLA-IS
PROJECT NUMBER: 3110-0925

MILITARY SPECIFICATION SHEET

TITLE
BEARING, BALL, AIRFRAME, ANTI-FRICTION, EXTRA LIGHT DUTY, PRECISION

SPECIFICATION SHEET NUMBER
MS21428 21 JUNE 95 REV C

SUPERSEDING
MS21428B 05 DEC 94

AMSC- N/A FSC 3110

DISTRIBUTION STATEMENT
A. Approved for public release, distribution is unlimited.

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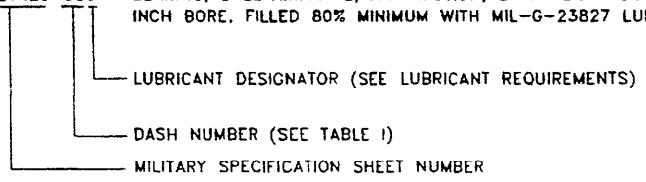
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REQUIREMENTS:

1. MATERIAL: RINGS: 52100 STEEL PER AMS6440 OR E52100 STEEL PER FED-STD-66.
   BALLS: 52100 STEEL PER AMS6440 OR 51100 STEEL PER AMS6449 OR E51100 OR E52100 STEEL PER FED-STD-66
   SEALS: POLYTETRAFLUOROETHYLENE PER AMS3652 OR POLYTETRAFLUOROETHYLENE SHEET, GLASS FABRIC REINFORCED PER AMS3666.
2. LUBRICANT: MIL-G-81322 OR MIL-G-23827. ALL BEARINGS SHALL BE PACKED WITH GREASE CONFORMING TO MIL-G-81322 UNLESS OTHERWISE SPECIFIED. IF MIL-G-23827 IS REQUIRED, ADD THE LETTER "G" AFTER THE MS21428 DASH NUMBER. MIL-G-23827 SHALL NOT BE USED FOR OPERATION WHERE TEMPERATURES EXCEED 250°F
3. HARDNESS: HEAT TREAT RINGS AND BALLS TO ROCKWELL "C" 60 TO 66 AND STABILIZE FOR OPERATION AT 250°F.
4. SURFACE ROUGHNESS: RACEWAYS AND BALLS SHALL BE 8 MICROINCHES Ra PER ANSI/ASME B46.1.
5. (C) PLATING: ALL EXTERNAL SURFACES EXCEPT BORE, SEALS AND SEAL RETAINERS SHALL BE ZINC-NICKEL IN ACCORDANCE WITH AMS 2417, TYPE 2, OR CADMIUM PLATED IN ACCORDANCE WITH QQ-P-416, TYPE I, CLASS 2, WITH A MINIMUM THICKNESS OF .0003 TO .0006 INCHES.
6. RADIAL INTERNAL CLEARANCE: .0001 TO .0005
7. RADIAL ECCENTRICITY: INNER RING: .0008 MAXIMUM. OUTER RING .0008 MAX.
8. LATERAL ECCENTRICITY: INNER RING: .0008 MAXIMUM. OUTER RING .0008 MAX.
9. PART NUMBER: THE PART NUMBER SHALL BE SEQUENCED LEFT TO RIGHT WITH DESIGNATIONS IN THE FOLLOWING SEQUENCE:

   EXAMPLE OF PART NUMBER.
   MS21428-38G = BEARING, BALL AIRFRAME, ANTIFRICTION, EXTRA LIGHT DUTY, PRECISION, .6250. INCH BORE, FILLED 80% MINIMUM WITH MIL-G-23827 LUBRICANT, 250°F MAX.

10. REMOVABLE SEALS ARE REQUIRED.

NOTES:
1. UNLESS OTHERWISE SPECIFIED, ALL DIMENSIONS ARE IN INCHES.
2. THIS SPECIFICATION TAKES PRECEDENCE OVER ANY OTHER DOCUMENTS REFERENCED HEREIN.
3. REFERENCED DOCUMENTS ARE THE ISSUES IN EFFECT AT THE DATE OF INVITATION FOR BID.

PREPARING ACTIVITY: NAVY-AS CUSTODIANS: ARMY-AV NAVY-AS AIR FORCE-99 DLA- REVIEWARMY-AR,AT,MI AIR FORCE-84, DLA-IS PROJECT NUMBER: 3110-0925	MILITARY SPECIFICATION SHEET TITLE BEARING, BALL, AIRFRAME, ANTI-FRICTION, EXTRA LIGHT DUTY, PRECISION	SPECIFICATION SHEET NUMBER MS21428 21 JUNE 95 REV C SUPERSEDING MS21428B 05 DEC 94 AMSC- N/A FSC 3110
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